



Evaluating Model Agreement in Explainable Fake News Detection using Transformer-Based Models

Measuring not just what transformer models predict — but how consistently, fairly, and explainably they agree.

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01 Agenda

What this defense will cover

01



Introduction

02



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**Research Gaps & Problem
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**Comparative Results &
Explainability**

07



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Reflections & Acknowledgements

Why fake news detection needs more than accuracy

How misinformation outpaces the truth



Social Media



News Websites



Content Apps

Spreads FASTER than the truth

Most impactful during:



Public Health



Elections



Finance



Governance

Transformers (BERT, RoBERTa, DistilBERT) detect fake news with high accuracy — but act as a “black box”, lacking explainability.



RESEARCH GAP

No existing framework combines transformer-based detection, explainability, and agreement evaluation in one holistic study.



AIM

Propose and evaluate an explainable, reliable framework for fake news detection using transformer-based models and agreement metrics — Cohen's Kappa and Fleiss' Kappa.

Literature Review

Key insights that shaped this research



Raza & Schwartz, 2025

Transformer models (BERT, RoBERTa, DistilBERT) outperform traditional ML in fake news detection.



Chaudhry et al., 2025

SHAP, LIME & attention visualization improve interpretability — no single method dominates.



Speith & Langer, 2023

Most papers optimize predictive performance but ignore agreement & reliability assessment.



Tajrian et al., 2023

Misinformation is socio-technical — driven by human bias and platform dynamics, not just text.



Kowald et al., 2024

Fairness and trustworthiness perspectives remain under-explored in detection models.



Wang, 2017

The LIAR benchmark dataset enables consistent comparison across fake-news studies.

Research Gaps & Challenges

Four unresolved issues in current fake-news detection literature



Limited Cross-Model Comparison

Few papers compare transformer explanation techniques against other algorithm families under one setup.



Inconsistent Preprocessing

Text preprocessing varies widely across studies, affecting both detection accuracy and explanation quality.



Understudied Agreement & Stability

Agreement between models — and explanation stability as sample size grows — is rarely examined.



Weak Generalizability Evidence

Transformer performance on heterogeneous, multi-domain datasets is seldom explored.

Problem Statement

Accuracy alone does not guarantee trust



Core Problem

Transformer models detect fake news with excellent accuracy — yet their predictions are not always explainable, undermining transparency.



Transparency Issue

Predictions lack human-interpretable reasoning, reducing trust in sensitive applications.



Adoption Risk

Systems shaping public opinion are hard to adopt without explainability.



Metric Shortcoming

Accuracy, Precision, Recall & F1 ignore interpretability entirely.



Explanation Consistency

No standard method exists to measure agreement between explanations.



Requirement: Fake-news detection models that deliver both strong classification results AND consistent, measurable explanations.

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Research Objectives

Three objectives guiding the study

01



Build & Compare Models

Train BERT, RoBERTa & DistilBERT for fake-news detection; benchmark against Logistic Regression, Random Forest, SVM & XGBoost baselines.

02



Explain Model Behaviour

Apply SHAP, LIME and BertViz attention visualization to interpret transformer decisions.

03



Measure Agreement & Fairness

Use Cohen's & Fleiss' Kappa for inter-model agreement; assess fairness via demographic parity and equalized odds.



1



Unified Transformer Comparison

Tested BERT, RoBERTa & DistilBERT under one experimental setup on the LIAR dataset.

2



Combined Explainability Methods

Merged SHAP, LIME & attention visualization into a single framework — prior work used only one.

3



Consistency Measurement

Applied Cohen's & Fleiss' Kappa alongside performance metrics to expose agreement issues.

4



Responsible-AI Evaluation

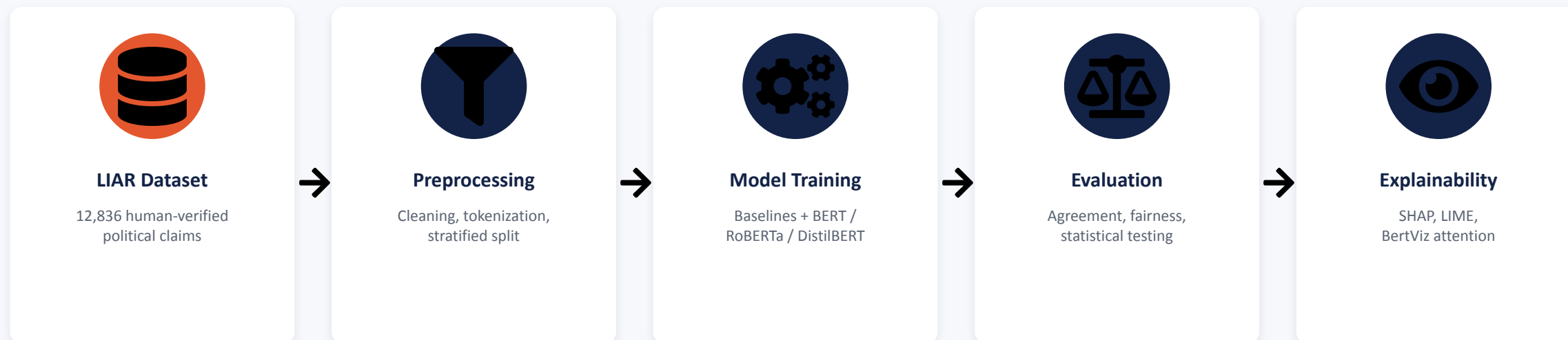
Linked explainability and reliability to trustworthiness, aligned with responsible-AI frameworks.

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Extensible Framework

Delivered a framework future researchers can extend for transparency-based misinformation detection.



KEY DATASET ATTRIBUTES (Wang, 2017)

- ✓ 12,836 statements sourced from the PolitiFact fact-checking site
- ✓ Chosen for annotation quality, prior-study prevalence & balanced true/false ratio
- ✓ Stratified split into train / validation / test sets after normalization & tokenization
- ✓ Baselines use TF-IDF features; transformers use their native tokenizers

09 Data Preprocessing

From raw claims to model-ready inputs



Text Cleaning & Normalization

- ✓ Removed formatting inconsistencies
- ✓ Preserved semantic meaning
- ✓ Handled missing / incomplete records



Dataset Splitting

- ✓ Stratified sampling applied
- ✓ Train / Validation / Test sets
- ✓ Balanced class distribution



Feature Preparation



Baseline Models

TF-IDF Vectorization

Converts textual statements into numerical feature vectors for Logistic Regression, SVM, Random Forest & XGBoost.



Transformer Models

Native Tokenization

BERT, RoBERTa & DistilBERT tokenizers prepare contextual inputs directly from raw claim text.



Development Environment

Python + Google Colab (cloud GPUs) • PyTorch • Hugging Face Transformers • Scikit-learn • SHAP • LIME • BertViz • Pandas • NumPy • Matplotlib

Baseline ML Models

Trained on TF-IDF features to benchmark performance and expose the limits of non-contextual representations.



Logistic Regression



Support Vector Machine



Random Forest



XGBoost

Transformer Models

Fine-tuned contextual language models — the core comparison group of this research.



BERT



RoBERTa



DistilBERT

Evaluation Framework

Beyond accuracy — four evaluation pillars



Classification Performance

Accuracy, Precision, Recall, F1-score & ROC-AUC — transformers outperform baselines, matching benchmarks like GETE & HEMT-Fake.



Agreement Measures

Cohen's Kappa (predictions vs. labels) and Fleiss' Kappa (BERT vs. RoBERTa vs. DistilBERT) fill the agreement-evaluation gap.



Statistical Testing

ANOVA confirms whether performance differences between models are statistically significant.

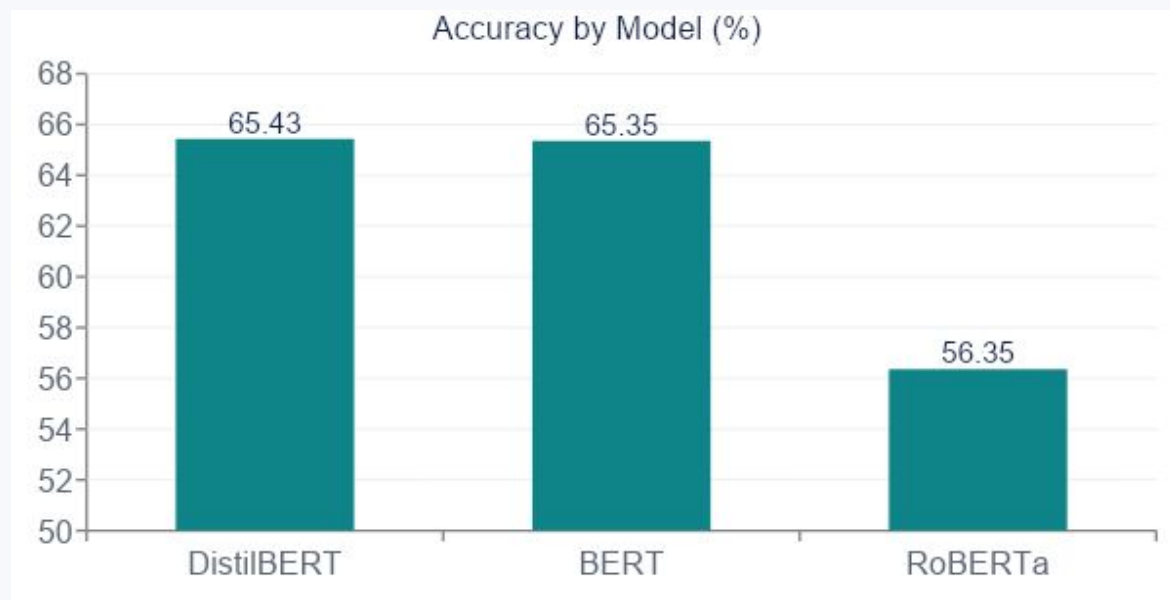


Fairness Criteria

Demographic Parity & Equalized Odds assess fair treatment across political groups, per trustworthy-AI principles.

Results and Analysis

Transformer performance on the LIAR benchmark



Model	Accuracy	Macro-F1	AUC-ROC
DistilBERT	65.43%	0.6448	0.6830
BERT	65.35%	0.6360	0.6914
RoBERTa	56.35%	0.3604	0.5199

Key Findings

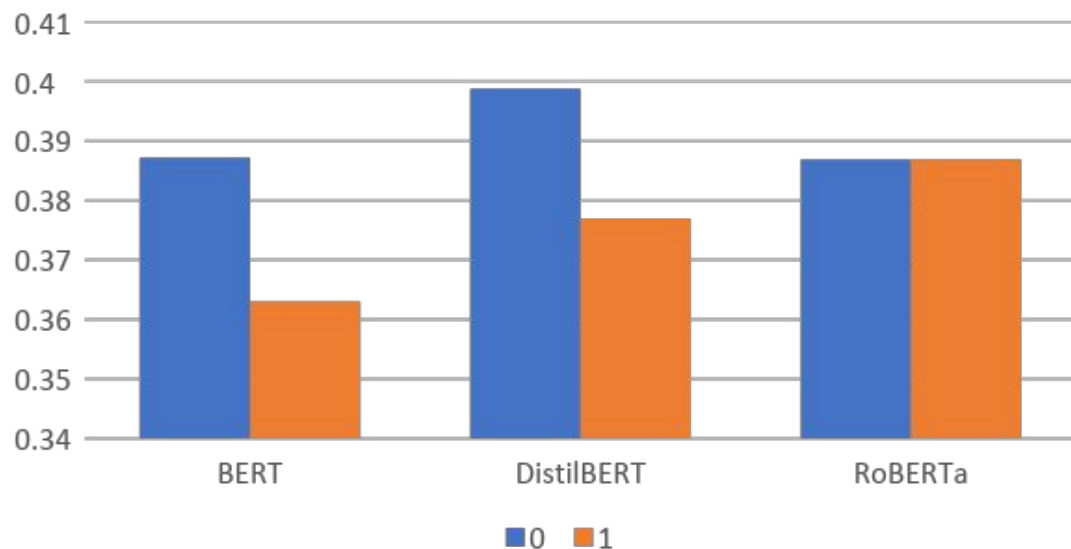
- ✓ DistilBERT slightly outperformed BERT despite being smaller — compression did not hurt performance.
- ✓ RoBERTa underperformed, likely due to sensitivity to class imbalance and short-statement context.
- ✓ Agreement stayed low: Cohen's Kappa 0.2974 (DistilBERT), 0.2730 (BERT) — below the 0.61 threshold for substantial agreement.
- ✓ ANOVA found no significant confidence difference ($F = 0.3396$, $p = 0.7122$).
- ✓ DistilBERT led on accuracy but showed the largest demographic disparity (0.1552); XGBoost was most balanced (0.0595).

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ANOVA RESULTS

Transformer performance on the LIAR benchmark

TWO – WAY SUMMARY (MODEL X CORRECT):



CORRECT MODEL	0	1
BERT	0.387159	0.363004
DistilBERT	0.398810	0.376928
RoBERTa	0.386867	0.386867

Key Findings

★ ONE – WAY ANOVA ON PREDICTION PROBABILITIES

✓ N = 200 (PER GROUP)

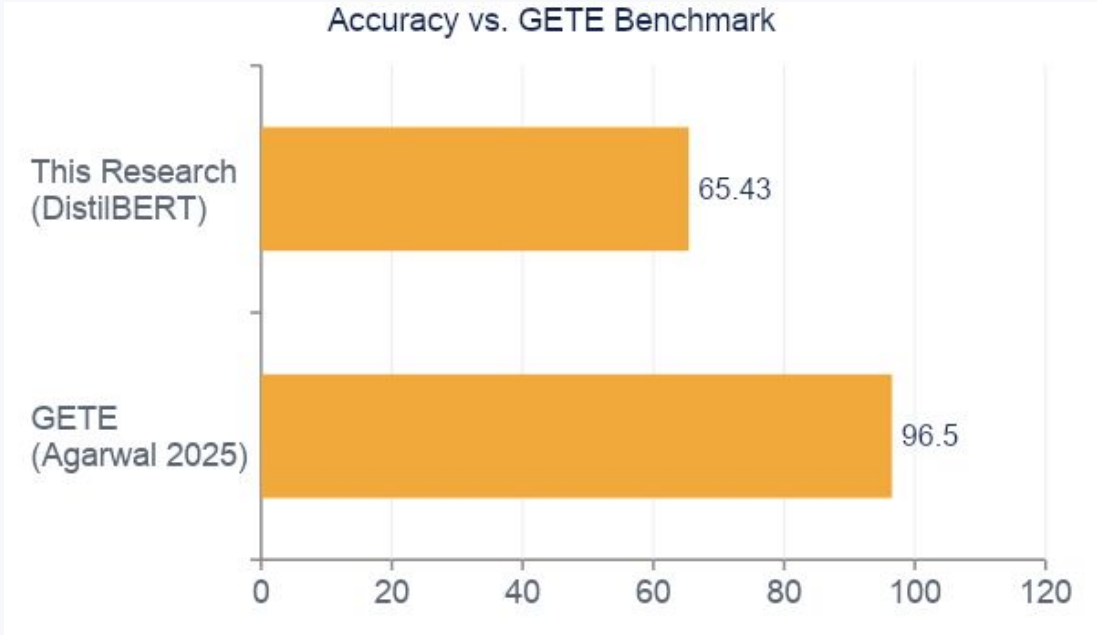
✓ F – STATISTIC = 0.2511

✓ P – VALUE = 0.7780

✓ SIGNIFICANT = NO

13 Comparison with Existing Work

Positioning this research against state-of-the-art systems



HEMT-Fake (Jadhav et al., 2025) reports high multimodal accuracy but is not directly numeric-comparable (different data modality).

Study	Approach	Accuracy
GETE (Agarwal et al., 2025)	Graph-Enhanced Transformer Ensemble	96.5%
HEMT-Fake (Jadhav et al., 2025)	Multimodal Transformer + XAI	High (multimodal)
This Research	DistilBERT + Agreement + Explainability	65.43%



Why the Gap Exists

- GETE/HEMT-Fake use graph-enhanced or multimodal architectures beyond this project's scope.
- This research targets trustworthiness — agreement, fairness & explainability — not maximum accuracy alone.
- Differences also stem from preprocessing, hyperparameter tuning & dataset choices.

14 Explainability Findings

SHAP, LIME & attention visualization compared

Method	Scope	Key Observation	Limitation
SHAP	Local & Global	Flagged politically-charged / emotionally-loaded tokens as high-importance	Sometimes weighted semantically weak stop-words
LIME	Local	Accessible, instance-specific; correctly flagged core claim terms	Variable across repeated runs (perturbation-based)
Attention Visualization	Internal / Layer-wise	Deeper layers focused on named entities & negation	Attention $\neq$ causal feature importance



Convergence on Correct Classifications

SHAP and LIME frequently agreed on salient tokens when the model classified correctly.



Divergence on False Classifications

More disagreement between explanation techniques on misclassifications — consistent with the low Cohen's Kappa. No single technique is universally necessary (Ribeiro et al. 2016; Lundberg & Lee 2017; Chaudhry et al. 2025).

Agreement, Reliability & Outcome Analysis

Kappa Agreement vs. Required Threshold ($\kappa \geq 0.61$)

0.2974

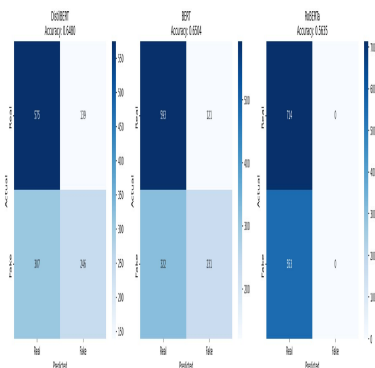
0.2730

Required 0.61

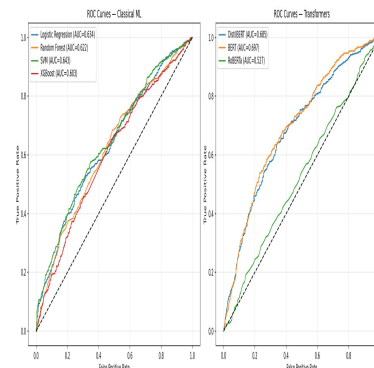
Conclusion: Good accuracy $\neq$ reliable decision-making.

16 Visual Results Gallery

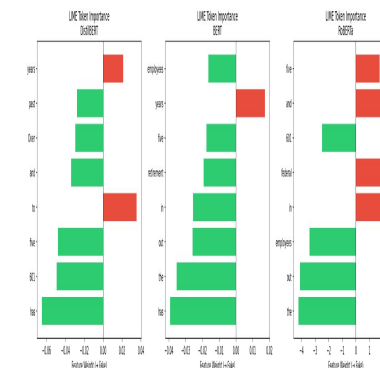
Performance and explainability outputs at a glance



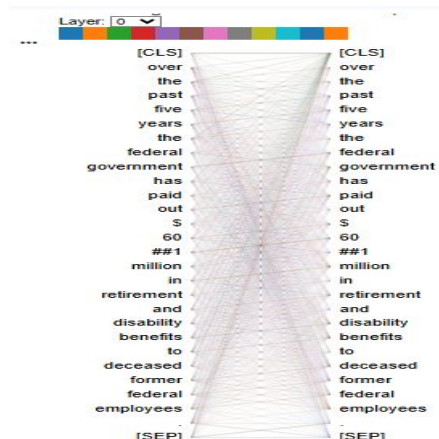
Confusion Matrix



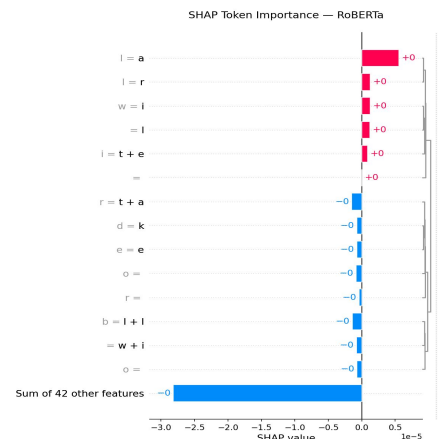
ROC-AUC Curve



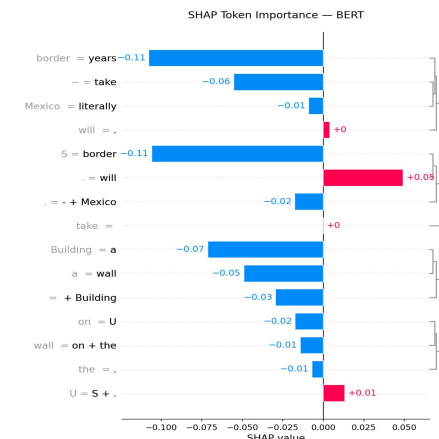
LIME Explanation



BertViz Attention



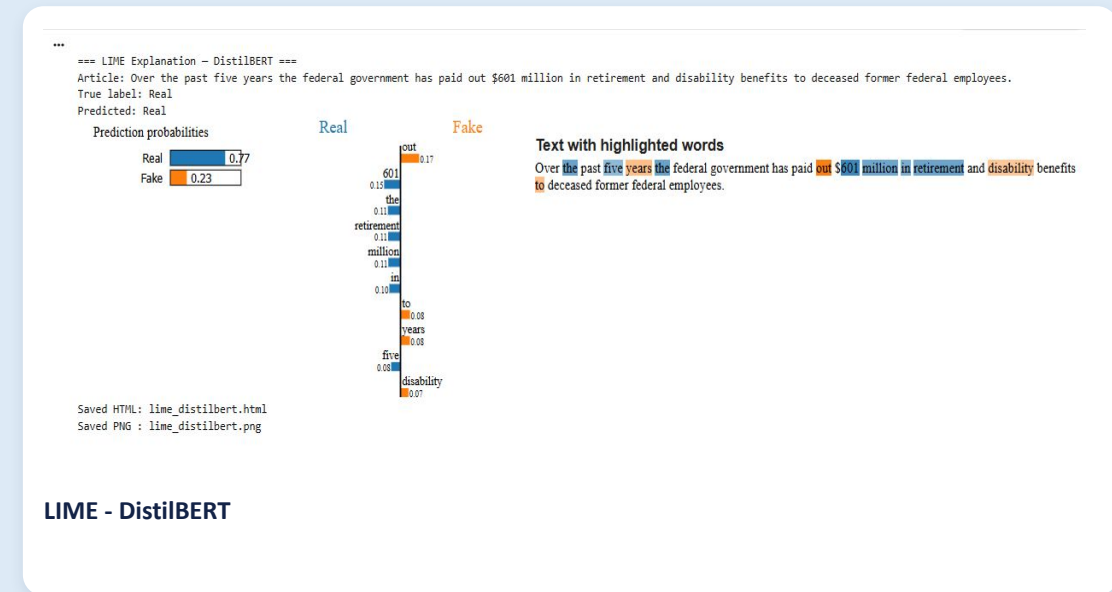
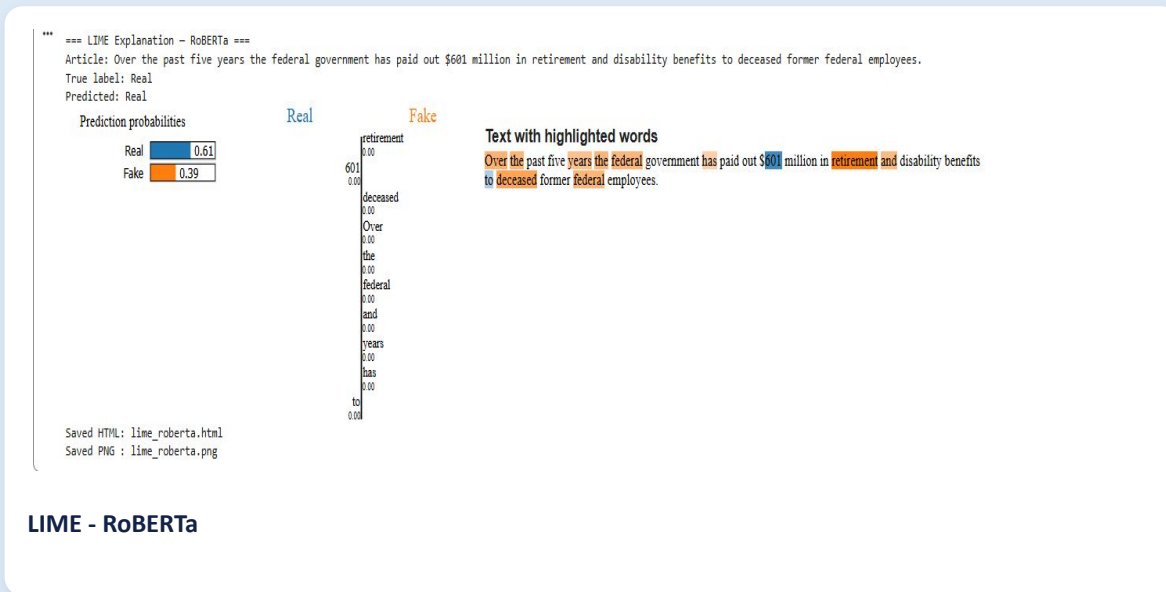
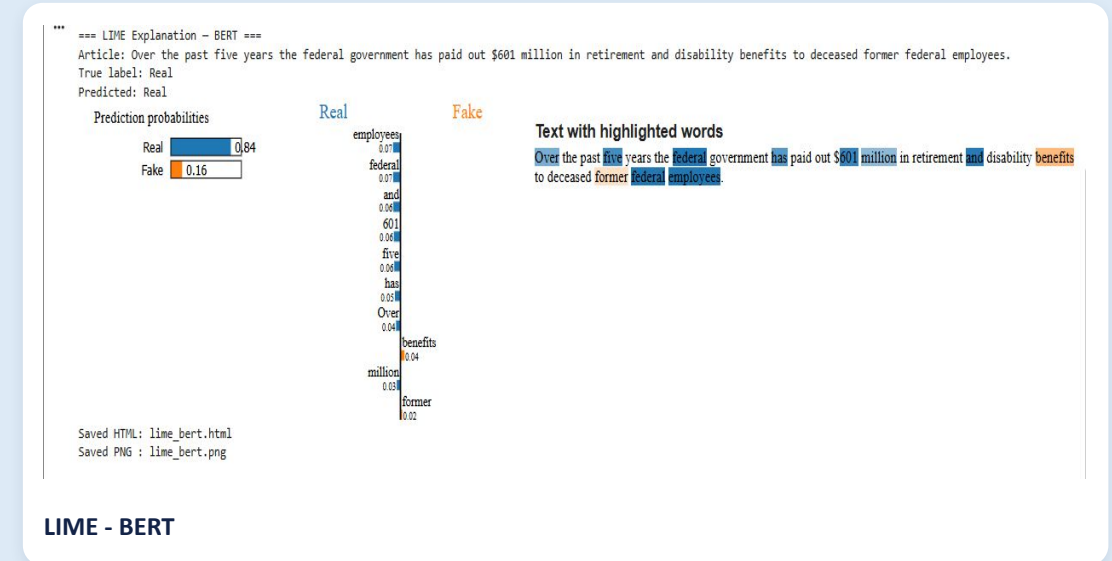
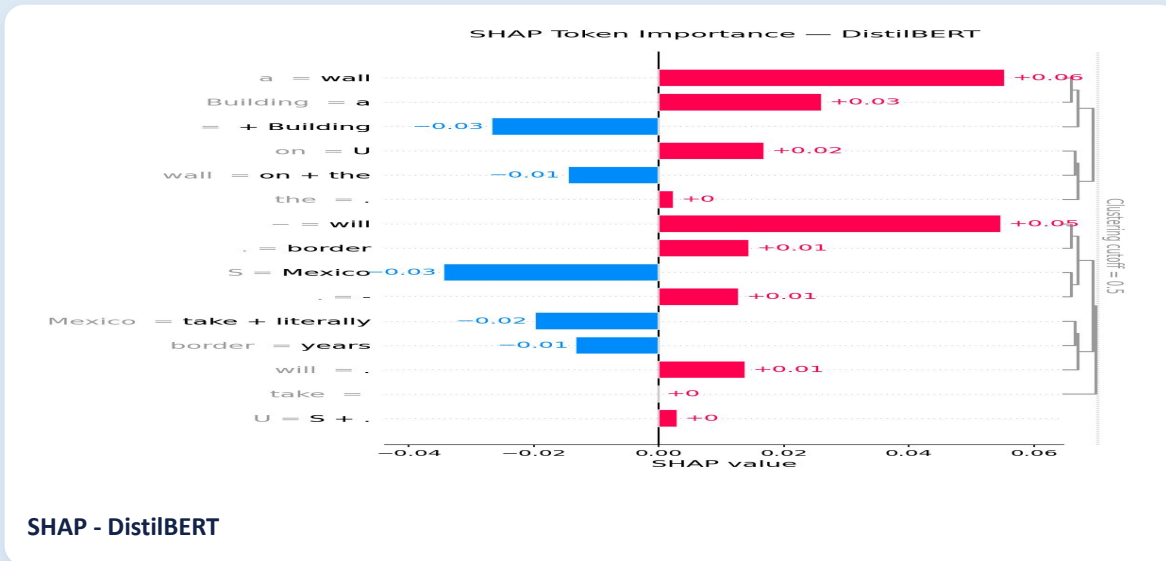
SHAP — Feature Impact



SHAP — Force Plot

16 Visual Results Gallery

Performance and explainability outputs at a glance



17 Conclusion & Future Work

This research built and evaluated an explainable, agreement-aware framework for transformer-based fake news detection — showing that strong accuracy does not guarantee explanation reliability or model consensus.

NEXT STEPS: FUTURE WORK



Larger, Domain-Tuned Models

Evaluate bigger transformer / recent LLM variants for stronger classification and explanation.



Multilingual & Cross-Domain Data

Extend beyond LIAR into health, financial and non-English misinformation.



More Reliable Explanations

Address inconsistency in SHAP, LIME & attention methods (Chaudhry et al., 2025).



Ensemble & Hybrid Models

Combine transformers with graph or multimodal models, as in GETE (Agarwal et al., 2025).



Extended Agreement Metrics

Broaden Cohen's & Fleiss' Kappa use for trustworthiness in automated fact-checking.